

A rank-one working model of behavior leakage

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1 Setup

We study *leakage*: a behavior trained into a model under one context (a system prompt, persona, or ICL prefix) also appearing under other, untrained contexts. The measured quantity is the trained–base log-probability of the behavior token(s) at the end of a bystander context’s own on-policy response.

Represent objects as residual-stream activation vectors at a fixed layer:

- a **context** c by $v_c \in \mathbb{R}^d$, read at the last prompt token (mean over a probe set of user questions);
- a **behavior** b by $v_b \in \mathbb{R}^d$, read at the position where the behavior is emitted (for the marker, the end-of-response slot).

The working premise is that, at that layer, the model’s context-to-behavior mapping is approximately linear,

$$v_b \approx Mv_c. \quad (1)$$

2 The update rule

Suppose context c' originally produces behavior b' , i.e. $v_{b'} = Mv_{c'}$, and we fine-tune the model to instead produce b'' under c' . The model says the new map is

$$M' = M + \frac{(v_{b''} - v_{b'}) v_{c'}^\top}{\|v_{c'}\|^2}, \quad (2)$$

a rank-one correction. The normalization makes the update fit the trained pair exactly: $M'v_{c'} = v_{b''}$. The induced change at any *bystander* context c is then

$$\Delta v_b(c) = \underbrace{(v_{b''} - v_{b'})}_{\text{fixed direction}} \cdot \underbrace{\frac{\langle v_{c'}, v_c \rangle}{\|v_{c'}\|^2}}_{\text{scalar coefficient}}. \quad (3)$$

Leakage to c is a single behavior direction, scaled by how much c overlaps the trained context.

3 Where the rule comes from

Three independent derivations land on the same shape.

(a) Delta rule on a linear map. One gradient step on $L = \frac{1}{2}\|v_{b''} - Mv_{c'}\|^2$ gives $\Delta M = \eta(v_{b''} - Mv_{c'}) v_{c'}^\top = \eta(v_{b''} - v_{b'}) v_{c'}^\top$. Equation (2) is this step with $\eta = 1/\|v_{c'}\|^2$ (the exact-fit step size). It is also the classical outer-product update of linear associative memories [1, 2]; in that literature, leakage is *crosstalk* between stored keys.

(b) Kernel view of fine-tuning. For a small parameter update $\delta\theta$, the function change at input x is $\Delta f(x) \approx \nabla_{\theta} f(x)^{\top} \delta\theta$. SGD on a single training input x' sets $\delta\theta \propto \nabla_{\theta} f(x') \times (\text{residual})$, so

$$\Delta f(x) \approx K(x, x') \times (\text{residual at } x'), \quad (4)$$

where K is the empirical neural tangent kernel [3]. Kernel dynamics have been shown to approximate language-model fine-tuning in several regimes [4]. Equation (3) is this statement with the activation inner product $\langle v_{c'}, v_c \rangle$ standing in for K , and $(v_{b''} - v_{b'})$ as the residual.

(c) LoRA is low-rank by construction. A LoRA update $\Delta W = BA$ has rank $\leq r$ [5], and training toward a single behavior tends to concentrate it in its top singular direction. The rank-one form is the $r=1$ idealization.

4 What it implies

Each structural feature of equation (3) is a falsifiable prediction.

P1 (shared direction).

The activation shift at the behavior read-out points in the *same* direction $(v_{b''} - v_{b'})$ for every bystander context; only the scalar coefficient varies. Operationally: stack per-bystander shift vectors, run an SVD, and the top component should dominate and match the pre-training behavior delta.

P2 (overlap coefficient).

The per-bystander coefficient is $\langle v_{c'}, v_c \rangle / \|v_{c'}\|^2$. This explains why context cosine similarity predicts leakage, and it is *asymmetric*: the coefficient from $c' \rightarrow c$ is $\cos \theta \cdot \|v_c\| / \|v_{c'}\|$, which is not invariant under swapping source and bystander. The norms carry the asymmetry.

P3 (residual scaling; behavior dependence).

The update magnitude scales with $\|v_{b''} - v_{b'}\|$, the gap between the trained behavior and what the source context already produces. If the source context already elicits the behavior (e.g. a system prompt instructing it), the residual is near zero and the rule predicts little leakage even when context similarity is high; a similarity-only predictor gets this case wrong. On the read-out side, the softmax compresses $\Delta \log P$ near saturation, so prior-dependence should appear in log-probability space and shrink in logit space.

P4 (additivity).

Training on two pairs sums their rank-one updates, so leakage from joint training should approximately equal the sum of the single-source leakages.

P5 (layer locality).

The relevant v_c is the one feeding the behavior's computation, so the most predictive layer should sit where the behavior is computed: late for a next-token marker, earlier for higher-level behaviors such as sycophancy.

P6 (cross-behavior transfer).

The same update moves a *different* behavior b at a fixed context in proportion to the projection of $(v_{b''} - v_{b'})$ onto b 's read-out direction, extending the rule from context generalization to behavior generalization.

5 How we are testing it

	Test	Status / result
P1	SVD of per-context activation shifts after a marker implant vs misalignment SFT (#521, controls #551)	Surprising split: misalignment SFT shifts all 14 probed contexts along one shared direction ($\cos 0.96\text{--}0.98$, $\sim$ half the spectrum; seed-stable); the marker implant does not, and the trained context is the <i>least</i> aligned. Contrast survives controls. Benign SFT also yields one shared direction (#552, running), so the geometry may be generic to plain SFT; a positive-only marker retrain (#561, running) tests whether the contrastive rig, not the behavior, sets the geometry.
P2	Correlate base-model context distances with measured leakage across source $\times$ bystander panels (#502, #489; sycophancy analogue)	Context similarity ranks marker leakage well ($ \rho \approx 0.76\text{--}0.79$ at layers 19–24) and extends to ICL/multi-turn contexts; on held-out persona panels geometry predicts the training-induced <i>change</i> but the structure does not transfer across panels (#553), and the base-model prior beats geometric distance on instruction contexts (#532). The magnitude half of P2 is unsettled in #521.
P3	Vary how strongly the source context already elicits the behavior; instructed-marker break case; track logit vs log-prob spaces (#500, #531, #532)	Growing support: the bystander’s own prior on the behavior is the most consistently supported predictor across panels; logit-space tracking is now in place to separate read-out compression from the update itself.
P4	Joint two-source training vs sum of singletons (#520, #527, #538)	Not yet diagnostic: additivity cosine reads 0.99, but each singleton’s shift matrix is itself near rank-one, so near-parallel constants sum trivially; the confound gates fail at every tested training depth. A regime where singleton shifts vary across contexts is needed.
P5	Layer sweep of predictor quality per behavior	Consistent: marker leakage is best predicted at layers 19–24; sycophancy at layers 1–8 (best cell: MMD at layer 7).
P6	Behavior-generalization testbed ($B \rightarrow B'$ leakage matrix, #545); context testbed (#537); formula-as-predictor (#510)	Queued: #537 running, #545 and #510 not yet run.

6 Known gaps

- The linear premise (1) is assumed, not fitted. A direct check is to regress v_b on v_c across many context–behavior pairs before training and measure how much variance a single linear map explains.
- The rule lives at one layer; training updates all layers. P5 is a heuristic patch, not a derivation.
- The P1 results carry a training-rig confound (contrastive marker-only loss vs plain SFT) until #561 lands.

- The read-out is nonlinear (softmax over the unembedding), so activation-space predictions must be compared against logits, with log-probabilities treated as a compressed view near saturation.

References

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